Interior angles of polygons

1

a 4 triangles

b 720°

2

a 4 triangles

b 720°

3

a

i 180 ii 180 iii 180

b 540

c $n-2$

d

i No ii No iii Yes iv No

4

	Number of Sides	Least Number of Triangles	Interior angle sum
Pentagon	5	3	540°
Hexagon	6	4	720°
Heptagon	7	5	900°
Octagon	8	6	1080°
Nonagon	9	7	1260°
Decagon	10	8	1440°

5

a 360° b 1800° c 2160° d 2520°

e 3240° f 6840°

6

a

i $y=63^{\circ}$ ii $x=64^{\circ}$

b

i $y=101^{\circ}$ ii $x=139^{\circ}$

7 $y=158^{\circ}$

8

a 1080° b 135°

9

a $x = 120^\circ$

b $x = 108^\circ$

10

a

i $n = 6$

ii Hexagon

b

i $n = 8$

ii Octagon

c

i $n = 3$

ii Triangle

11 Because then $n = 7.2$, and it is not possible to have 7.2 sides of a polygon.

12 1260°

13 135°

Exterior angles of polygons

14

a

i $x = 73^\circ$

ii $a = 88^\circ$

iii $b = 92^\circ$

iv $c = 107^\circ$

v $d = 73^\circ$

vi 360°

b The sum of exterior angles in a quadrilateral is 360°.

15 $y = 98^\circ$

16

a $x = 81^\circ$

b $x = 7^\circ$

17

a 72°

b 45°

c 120°

d 40°

18

a 120°

b 90°

c 144°

d 150°